

# A Morse-Theoretic Construction of the Atiyah–Hirzebruch Spectral Sequence

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## Contents

|                                                                  |           |
|------------------------------------------------------------------|-----------|
| <b>Notation &amp; Conventions</b>                                | <b>2</b>  |
| <b>1 Introduction</b>                                            | <b>6</b>  |
| <b>2 Vector Bundles and Riemannian Geometry</b>                  | <b>7</b>  |
| 2.1 Differentiable Structures on Topological Manifolds . . . . . | 7         |
| <b>3 Complex K Theory</b>                                        | <b>20</b> |
| 3.1 Mathematical Goals . . . . .                                 | 20        |
| 3.2 Classification of Vector Bundles . . . . .                   | 20        |
| <b>4 Orientations</b>                                            | <b>31</b> |
| 4.1 Mathematical Goals . . . . .                                 | 31        |
| <b>5 Morse Theory</b>                                            | <b>33</b> |
| <b>6 Title</b>                                                   | <b>36</b> |
| 6.1 Mathematical Goals . . . . .                                 | 36        |
| <b>7 Title</b>                                                   | <b>37</b> |
| 7.1 Mathematical Goals . . . . .                                 | 37        |

## Notation & Conventions

This page collects all notational conventions used in this thesis. It is to be **removed before submission**.

### 1. Manifolds, Maps, and the Differential

| Symbol                        | Meaning & Convention                                                                                                                                                                                                                |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $M, N$                        | smooth manifolds (always assumed without boundary unless stated)                                                                                                                                                                    |
| $T_p M, T_p^* M$              | tangent / cotangent space at $p \in M$                                                                                                                                                                                              |
| $TM, T^* M$                   | tangent / cotangent bundle                                                                                                                                                                                                          |
| $df$                          | exterior differential of $f: M \rightarrow \mathbb{R}$ ; a 1-form on $M$                                                                                                                                                            |
| $d_p f = (df)_p$              | differential of $f$ at the point $p$ ; a linear map $T_p M \rightarrow \mathbb{R}$ .<br><b>Rule:</b> write $d_p f$ (macro <code>\dat{p}f</code> ) when evaluating at a specific point. Write $df$ for the full 1-form / bundle map. |
| $d_p F$                       | for $F: M \rightarrow N$ : the linear map $T_p M \rightarrow T_{F(p)} N$ . Also written $(dF)_p$ ; never write $\frac{d}{dx} F$ for a map between manifolds.                                                                        |
| $\frac{d}{dt}$                | total derivative w.r.t. a <i>single</i> real variable $t$ (roman d, macro <code>\diff</code> ). Use for curves, flows evaluated on a trajectory.                                                                                    |
| $\frac{\partial}{\partial t}$ | partial derivative; use only when <i>at least two independent</i> variables/parameters are present. <b>Never</b> write $\partial/\partial t$ for the time-derivative along a flow line.                                             |
| $\nabla_v f$                  | covariant derivative of $f$ in direction $v$ (Levi-Civita or chosen connection)                                                                                                                                                     |
| $\text{grad}_g f$             | gradient of $f$ with respect to Riemannian metric $g$ ; characterised by $g(\text{grad}_g f, -) = df$                                                                                                                               |

### 2. Flows and Dynamical Systems

| Symbol                      | Meaning & Convention                                                                                                                                                                                                         |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\varphi_t(x)$              | flow of a vector field at time $t$ through initial point $x$ . <b>Convention:</b> always write $\varphi_t$ , not $\varphi(t, \cdot)$ ; think of $\varphi_t \in \text{Diff}(M)$ as a one-parameter family of diffeomorphisms. |
| $\varphi_t = e^{tX}$        | flow of the vector field $X$ ; $\varphi_0 = \text{id}$ .                                                                                                                                                                     |
| $\frac{d}{dt} \varphi_t(x)$ | time-derivative of the flow line $t \mapsto \varphi_t(x)$ ; equals $X(\varphi_t(x))$ . Use roman d, never $\partial$ .                                                                                                       |
| $W^u(p), W^s(p)$            | unstable / stable manifold of a critical (rest) point $p$ (macros <code>\Wu{p}</code> , <code>\Ws{p}</code> )                                                                                                                |

| Symbol              | Meaning & Convention                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------|
| $\mathcal{M}(p, q)$ | moduli space of (unparametrised) flow lines from $p$ to $q$<br>(macro <code>\Mflow{p}{q}</code> ) |

### 3. Restrictions and Evaluations

**Core rule:** distinguish carefully between a *fiber*, a *restriction to a subspace*, and a *value/evaluation*.

| Symbol                 | Meaning & Convention                                                                                                                                                                              |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E_p$                  | <b>fiber</b> of a vector bundle $E \rightarrow X$ over $p \in X$ . <b>Preferred over</b> $E _{\{p\}}$ for fibers; use macro <code>E_p</code> .                                                    |
| $E _A$                 | <b>restriction</b> of the bundle $E$ to the subspace $A \subseteq X$ ; this is itself a bundle over $A$ . Macro: <code>\restr{E}{A}</code> .                                                      |
| $\xi _A$               | section $\xi \in \Gamma(E)$ restricted to $A$ ; yields $\xi _A \in \Gamma(E _A)$ .                                                                                                                |
| $\xi(p)$ or $\xi_p$    | <b>value</b> of the section $\xi$ at the point $p$ ; an element of $E_p$ . <b>Avoid</b> $\xi _p$ for this; use $\xi(p)$ .                                                                         |
| $\omega_p = \omega _p$ | value of a differential $k$ -form $\omega$ at $p$ ; an element of $\Lambda^k T_p^* M$ . Acceptable to write $\omega _p$ only when the point-evaluation is being <i>emphasised</i> over the fiber. |
| <b>Summary</b>         | $E_p$ = fiber (bundle); $E _A$ = restricted bundle; $\xi(p)$ = value of section; $\omega_p$ = value of form.                                                                                      |

### 5. Chain Complexes and Homology Theories

Three homology theories appear simultaneously; they are always decorated.

| Theory   | Chain complex                        | Boundary / Homology                                                                                                              |
|----------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Singular | $({}^S C_*(X), {}^S \partial)$       | boundary ${}^S \partial$ ; homology $H^S H_*(X)$ .<br>Macros: <code>\ChS</code> , <code>\bS</code> , <code>\HS</code> .          |
| CW       | $({}^{CW} C_*(X), {}^{CW} \partial)$ | boundary ${}^{CW} \partial$ ; homology $H^{CW} H_*(X)$ .<br>Macros: <code>\ChCW</code> , <code>\bCW</code> , <code>\HCW</code> . |
| Morse    | $({}^M C_*(f), {}^M \partial)$       | boundary ${}^M \partial$ ; homology ${}^M H_*(f)$ .<br>Macros: <code>\ChM</code> , <code>\bM</code> , <code>\HM</code> .         |

**Connecting homomorphisms in long exact sequences:**  $: {}^M \delta, {}^{CW} \delta, {}^S \delta$  with the macros: `\connM`, `\connCW`, `\connS`.

**Co-homology:** For co-homology we use upper indices on the co-boundary maps:  ${}^M d, {}^{CW} d, {}^S d$  with the macros: `\cbM`, `\cbCW`, `\cbS`. And for the co-homology we use the homologie macros with upper indices.

| Theory                                                                                                                 | Chain complex | Boundary / Homology |
|------------------------------------------------------------------------------------------------------------------------|---------------|---------------------|
| <b>Filtration:</b> $F^{p\mathcal{S}}C_*(X)$ for the filtration of the singular complex; boundary is still $\partial$ . |               |                     |

## 6. Vector Bundles and K-Theory

| Symbol                    | Meaning                                                                                          |
|---------------------------|--------------------------------------------------------------------------------------------------|
| $\text{Vect}(X)$          | set of isomorphism classes of complex vector bundles over $X$<br>(macro <code>\Kvect{X}</code> ) |
| $[E]$                     | stable isomorphism class / element in $K(X)$                                                     |
| $K(X) = KU(X)$            | complex K-theory (default); macro <code>\K</code> , <code>\KU</code>                             |
| $KO(X)$                   | real K-theory; macro <code>\KO</code>                                                            |
| $\tilde{K}(X)$            | reduced complex K-theory                                                                         |
| $E \oplus F, E \otimes F$ | Whitney sum / tensor product of bundles                                                          |
| $E _A$                    | restriction of bundle $E \rightarrow X$ to subspace $A$ (same convention as §3)                  |
| $E_p$                     | fiber of $E$ over $p$ (same convention as §3)                                                    |

## 7. Spectral Sequences

| Symbol           | Meaning                                                                                                                                 |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| $E_{p,q}^r$      | $(p, q)$ -entry on the $r$ -th page of a spectral sequence; macro <code>\Epq{r}{p,q}</code>                                             |
| $d^r$            | differential on the $r$ -th page; $d^r: E_{p,q}^r \rightarrow E_{p-r, q+r-1}^r$ (cohomological convention); macro <code>\drpq{r}</code> |
| AHSS             | Atiyah–Hirzebruch spectral sequence: $E_{p,q}^2 \cong H^p(X; K^q(\text{pt})) \Rightarrow K^{p+q}(X)$                                    |
| $E_{p,q}^\infty$ | abutment / $E_\infty$ -page                                                                                                             |

## 8. Frequently Used Macros (Quick Reference)

| Macro                     | Output   | Use                            |
|---------------------------|----------|--------------------------------|
| <code>\diff</code>        | $d$      | exterior deriv. / total deriv. |
| <code>\dat{p}</code>      | $d_p$    | deriv. at point $p$            |
| <code>\restr{E}{A}</code> | $E _A$   | restriction                    |
| <code>\Wu{p}</code>       | $W^u(p)$ | unstable manifold              |

| Macro                     | Output                | Use                            |
|---------------------------|-----------------------|--------------------------------|
| <code>\Ws{p}</code>       | $W^s(p)$              | stable manifold                |
| <code>\Mflow{p}{q}</code> | $\mathcal{M}(p, q)$   | moduli of flow lines           |
| <code>\ChM</code>         | $\mathcal{M}C_*$      | Morse chain complex            |
| <code>\bM</code>          | $\mathcal{M}\partial$ | Morse boundary                 |
| <code>\HM</code>          | $\mathcal{M}H_*$      | Morse homology                 |
| <code>\ChCW</code>        | ${}^{CW}C_*$          | CW chain complex               |
| <code>\bCW</code>         | ${}^{CW}\partial$     | CW boundary                    |
| <code>\HCW</code>         | $H^{CW}H_*$           | CW homology                    |
| <code>\conn</code>        | $\delta$              | connecting homom. ( $\delta$ ) |
| <code>\K</code>           | $K$                   | K-theory                       |
| <code>\KO</code>          | $KO$                  | real K-theory                  |
| <code>\KU</code>          | $KU$                  | complex K-theory               |
| <code>\Epq{r}{p,q}</code> | $E_{p,q}^r$           | SS entry                       |
| <code>\drpq{r}</code>     | $d^r$                 | SS differential                |
| <code>\Crit</code>        | $\text{Crit}$         | critical point set             |
| <code>\ind</code>         | $\text{ind}$          | Morse index                    |
| <code>\grad</code>        | $\text{grad}$         | gradient                       |
| <code>\euler</code>       | $e$                   | Euler number (roman e)         |
| <code>\ii</code>          | $i$                   | imaginary unit (roman i)       |

## 1 Introduction

## 2 Vector Bundles and Riemannian Geometry

sec:background-  
d-geometry

Due to differential geometry and the theory of vector bundles being areas of mathematics used by pure mathematicians interested in analytical aspects like curvature, by topologists interested in invariants like Euler characteristics, physicists using the spaces as playgrounds for general relativity theories, and many others, there are numerous competing notational conventions. This thesis lies at the tripoint of differential geometry, dynamical systems, and algebraic topology. Hence, there is no a priori preferred notational convention. In this chapter, we establish the necessary background in the theory of vector bundles and Riemannian manifolds whilst not proving all the basics; we establish the notation used throughout this thesis. We establish manifolds and vector bundles in parallel, as they are interdependent on one another.

### 2.1 Differentiable Structures on Topological Manifolds

We begin by defining a topological manifold. Afterwards, we equip that space with with increasingly more structure and hence reduce the generality until we arrive at the playground of the thesis:

def:vb-topolo-  
gicalManifold

**Definition 2.1.1** (Topological Manifold). A second countable Hausdorff space  $M$  is called a **topological manifold** of dimension  $m \in \mathbb{N}$ , if it is locally homeomorphic to  $\mathbb{R}^m$ . To be precise, for all  $p \in M$  there exists an open neighbourhood  $U \subseteq M$  of  $p$ , an open set  $V \subseteq \mathbb{R}^m$  and a map  $\varphi : U \rightarrow V$  that is a homeomorphism. We call the map  $\varphi : U \rightarrow V$  a **chart around  $p$  on  $M$** , and  $\varphi^{-1}$  a **local coordinate system around  $p$  on  $M$** .

def:vb-differ-  
entiableManif-  
old

**Definition 2.1.2** (Differentiable Manifold). Let  $M$  be a topological manifold of dimension  $m$ .

1. A **differentiable atlas of class  $r \in \mathbb{N} \cup \{\infty\}$**  is a family of charts  $\mathfrak{A} = (\varphi_i : U_i \rightarrow V_i)_{i \in I}$  such that
  - a)  $\bigcup_{i \in I} U_i = M$ , meaning that  $(U_i)$  is an open covering of  $M$ .
  - b) For every pair  $(i, j) \in I^2$ , the **transition function**:

$$\begin{aligned} \varphi_{ij} : \varphi_j(U_i \cap U_j) &\rightarrow \varphi_i(U_i \cap U_j) \\ x &\mapsto (\varphi_i \circ \varphi_j^{-1})(x) \end{aligned}$$

is differentiable of class  $r$ .

We call such an atlas a  $C^r$ -atlas.

2. Two  $C^r$ -atlases  $\mathfrak{A}$  and  $\mathfrak{B}$  are called **equivalent** if the family  $\mathfrak{A} + \mathfrak{B} = (\varphi_i, \varphi_j)_{ij}$  is a  $C^r$ -atlas.

A **differentiable structure of class  $r$**  on  $M$  is an equivalence class  $c$  of  $C^r$ -atlases. For  $r = \infty$ , we call the pair  $(M, c)$  a smooth manifold.

**Corollary 2.1.3.** Every transition function  $\varphi_{ij}$ ,  $i, j \in I^2$  of a differentiable atlas  $\mathfrak{A} = (\varphi_i)_{i \in I}$  is not merely a homeomorphism but also a diffeomorphism. This follows from the fact that  $\varphi_{ij}^{-1} = \varphi_{ji}$  which provides smooth inverses.

def:vb-smooth  
Functions

**Definition 2.1.4** (Smooth Functions). Let  $(M, c)$  be a differentiable manifold of class  $r$  and  $U \subseteq M$  open. We call a continuous function

$$f : U \rightarrow \mathbb{R}$$

**differentiable of class  $r$** , if for any chart (and hence for all)  $(\varphi_i)_{i \in I} = \mathfrak{A} \in c$  the compositions  $f \circ \varphi_i^{-1}$  are differentiable of class  $r$ . For  $r = \infty$  we define:

$$\mathcal{E}(U) = \{f : U \rightarrow \mathbb{R} \text{ continuous} \mid f \text{ is differentiable of class } \infty\}.$$

**Corollary 2.1.5.** Let  $(M, c)$  be a smooth manifold of dimension  $m$  and  $U \subseteq M$  be an open subset. With pointwise defined operations, the set  $(\mathcal{E}(U), +, \cdot, \circ)$  becomes an  $\mathbb{R}$ -algebra. Furthermore,  $\mathcal{E}$  becomes a sheaf of  $\mathbb{R}$ -algebras.

*Proof.* There is not really a need for a proof. However, it might help to work through the definition of a sheaf. First,  $\mathcal{E}$  is a presheaf, where the restriction in the domain of a function gives the needed restriction homomorphism:

$$\begin{aligned} \text{res}_V^U : \mathcal{E}(U) &\rightarrow \mathcal{E}(V) \\ f &\mapsto f|_V. \end{aligned}$$

The required properties of a presheaf are trivial. Furthermore, this gives a sheaf as the requirement of locality is trivial for functions and the property of gluing is also trivial for functions, since differentiability is a local property.  $\square$

**Definition 2.1.6.** If  $p \in M$  is fix,  $f \in \mathcal{E}(U)$  and  $g \in \mathcal{E}(U')$  such that  $p \in U \cap U'$  we say that  $f$  and  $g$  have the same **germ in  $p$** , if there is another open neighborhood  $W \subseteq U \cap U'$  of  $p$  such that  $f|_W = g|_W$ . This defines an equivalence relation  $\sim_p$ . An equivalence class  $s$  of local functions around  $p$  is called a **germ in  $p$** . We write  $s = f_p$ , if  $s = [f]$  with  $f \in \mathcal{E}(U)$ . We write

$$\mathcal{E}_p(M) = \left( \sum_{U \text{ open}, p \in U} \mathcal{E}(U) \right) / \sim_p$$

for the set of germs, and call it the **stalk in  $p$** . Where  $\sum$  denotes the co-product (disjoint union) in **Top**.

**Corollary 2.1.7.** For a smooth manifold  $(M, c)$  the set  $\mathcal{E}_p(M)$  inherits an  $\mathbb{R}$ -algebra structure from the  $\mathcal{E}(U)$ . Furthermore, it carries a natural (evaluation-)homomorphism:

$$\begin{aligned} \text{eval}_p : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto f(p) =: f_p(p) \end{aligned}$$

The stalks are also local rings with maximal ideal  $\mathfrak{m}_p = \ker(\text{eval}_p)$ . Hence, the pair  $(M, \mathcal{E})$  gives us a locally ringed space.



*Proof.* We only need to verify the local ring structure of the stalks. An element  $f_p \in \mathcal{E}_p(M)$  is invertible if and only if  $f(p) \neq 0$ , equivalently if  $f_p \notin \ker(\text{eval}_p)$ . Thus  $\mathcal{E}_p(M)/\ker(\text{eval}_p)$  is a field, which proves that  $\ker(\text{eval}_p)$  is maximal.  $\square$

**Definition 2.1.8.** Let  $(M, c)$  be a smooth manifold of dimension  $m$  and  $p \in M$ . We call an  $\mathbb{R}$ -linear map  $\delta : \mathcal{E}_p(M) \rightarrow \mathbb{R}$  a **derivation**, if it satisfies the Leibniz rule:

$$\delta(f_p \cdot g_p) = \delta(f_p)g_p(p) + f_p(p)\delta(g_p) \quad \text{for all } f_p, g_p \in \mathcal{E}_p(M).$$

We call  $\text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R})$  the set of derivations and give it an  $\mathbb{R}$ -vector space structure by pointwise operations. We define the **tangent space of  $M$  at  $p$**  to be the vector space

$$T_p M := \text{Der}_{\mathbb{R}}(\mathcal{E}_p(M), \mathbb{R}).$$

**Corollary 2.1.9.** Let  $(M, c)$  be a smooth manifold of dimension  $m$  and  $\varphi : U \rightarrow V$  be a chart around  $p$  with  $x_0 = \varphi(p)$  (where  $\varphi \in \mathfrak{A} \in c$ ). Then we define the derivation

$$\begin{aligned} \partial_i(p) : \mathcal{E}_p(M) &\rightarrow \mathbb{R} \\ f_p &\mapsto \partial_i(p)(f_p) := \frac{\partial(f \circ \varphi^{-1})}{\partial x_i} \Big|_{x_0}. \end{aligned}$$

This is well-defined and hence gives a tangent vector. Moreover, the family

$$(\partial_1(p), \dots, \partial_m(p))$$

defines a basis of  $T_p M$ . Hence, the dimension of  $T_p M$  is  $m$ .

This is a good point to take a break from smooth manifolds and introduce the structure of vector bundles.

Analogously to the definition of a smooth manifold, we start by defining families of vector spaces and equip them with increasingly more structure:

def:vb-vector  
Bundles

**Definition 2.1.10 (Vector Bundles).** A **complex vector bundle** of rank  $m$  is a surjection

$$\pi : E \rightarrow X$$

such that  $E$  and  $X$  are topological spaces and  $\pi$  is continuous. Furthermore  $\pi$  satisfies:

1. For all  $x \in X$ ,  $\pi^{-1}(x)$  carries the structure of an  $m$ -dimensional  $\mathbb{C}$ -vector space.
2. For all  $x \in X$ , there exist a neighbourhood  $U \subseteq X$  and a homeomorphism

$$\psi : E|_U := \pi^{-1}(U) \rightarrow U \times \mathbb{C}^m,$$

that is compatible in the following sense: Let  $\pi_1$  denote the projection  $U \times \mathbb{C}^m \rightarrow U$  onto the first factor. Then

$$\pi_1 \circ \psi = \pi$$

and

$$\psi_x : E_x := \pi^{-1}(x) \rightarrow \{x\} \times \mathbb{C}^m$$

is a vector space isomorphism.

We call  $E$  the **total space**,  $X$  the **base space**,  $E_x$  the **fibre over  $x$**  and  $\psi$  a **local trivialization**.

**Definition 2.1.11.** Let  $\pi : E \rightarrow X$  be a vector bundle. A continuous function  $s : X \rightarrow E$  such that  $\pi \circ s = \text{id}_X$  is called a (global) **section** (if  $s : U \subseteq X \rightarrow E$  we call it a local section). We denote the space of global sections by  $\Gamma(E)$  and by  $\Gamma(U)$  the space of local sections over an open subset  $U \subseteq X$ .

def:vb-homomorphismOfVectorBundles

**Definition 2.1.12** (Homomorphism of Vector Bundles). A **homomorphism between two vector bundles**  $\pi : E \rightarrow X$  and  $\tilde{\pi} : F \rightarrow X$  over the same base space is a continuous function  $\varphi : E \rightarrow F$  such that:

$$1. \quad \begin{array}{ccc} E & \xrightarrow{\varphi} & F \\ \downarrow \pi & \searrow \tilde{\pi} & \\ X & & \end{array} \quad \text{commutes.}$$

2. For all  $x \in X$  the function  $\varphi_x : E_x \rightarrow F_x$  is linear.

def:vb-pullbackOfVectorBundles

**Definition 2.1.13** (Pullback of Vector Bundles). Let  $\pi : E \rightarrow Y$  be a vector bundle,  $X$  be a topological space and  $f : X \rightarrow Y$  be continuous. Then we define the **pullback**  $f^*\pi : f^*(E) \rightarrow X$  as follows: The total space  $f^*(E)$  is defined as the subspace of  $(x, e) \in X \times E$  such that  $f(x) = \pi(e)$ , and the map  $f^*\pi$  is the projection to the second factor restricted to said subspace.

**Example 2.1.14.** Let  $\pi : E \rightarrow X$  be a vector bundle. For any  $Y \subseteq X$  we have the vector bundle

$$\pi|_Y : \pi^{-1}(Y) \rightarrow Y,$$

which is defined as the pullback along the inclusion of  $Y$  into  $X$ .

rem:vb-restrictionOfBundles

**Remark 2.1.15.** We will often omit the notation of the projection and denote a vector bundle by its total space. The restriction  $\pi|_Y : \pi^{-1}(Y) \rightarrow Y$  is often denoted by  $E|_Y$ . The restriction may equivalently be defined by restricting the domain and codomain of the projection.

rem:vg-pullbackInducesBundleMap

**Remark 2.1.16.** Notice that this construction of the pullback  $f^*\pi : f^*(E) \rightarrow X$  is the simplest space to make the square commute:

$$\begin{array}{ccc} f^*(E) & \xrightarrow{\pi_2} & E \\ \downarrow f^*\pi & & \downarrow \pi \\ X & \xrightarrow{f} & Y \end{array}$$

The notion of simplicity is captured by the universal property, that for any other vector bundle  $\tilde{\pi} : \tilde{E} \rightarrow X$  and function  $\tilde{f} : \tilde{E} \rightarrow E$  there exists a unique  $h : \tilde{E} \rightarrow f^*(E)$  (which is given by  $h(y) := (\tilde{\pi}(y), \tilde{f}(y))$ ) such that the following diagram commutes:

$$\begin{array}{ccccc}
\tilde{E} & & \xrightarrow{\tilde{f}} & & E \\
\downarrow h & & \searrow & & \downarrow \pi \\
& f^*(E) & \xrightarrow{\pi_2} & E & \\
& \downarrow f^*\pi & & \downarrow \pi & \\
& X & \xrightarrow{f} & Y & 
\end{array}$$

(Note: A curved arrow labeled  $\tilde{\pi}$  also points from  $\tilde{E}$  to  $X$ .)

**Remark 2.1.17.** Let  $\pi : E \rightarrow X$  be a vector bundle. Then the space of sections  $\Gamma(E)$  is a (in general, infinite-dimensional)  $\mathbb{R}$ -vector space. Furthermore, and more relevant to the purposes of this thesis, we view  $\Gamma(E)$  (and  $\Gamma(U)$  for open subsets) as a  $C(X)$  (or  $C(U)$ ) module. The latter denotes the ring of continuous functions from  $E$  (or  $U$ ) to  $\mathbb{R}$ .

**Example 2.1.18.** For any smooth manifold  $M$  of dimension  $m$ , we can equip the union

$$TM := \bigsqcup_{p \in M} T_p M$$

with a topology, such that  $\pi : TM \rightarrow M$  defines a vector bundle which we call **tangent bundle**. For this we first define the trivialisations and then equip  $TM$  with a topology to make them homeomorphisms: Let  $\varphi : U \rightarrow V$  be a chart on  $M$ . If we denote the  $i$ -th coordinate function of  $\varphi$  by  $\varphi_i$  we can define the bijection

$$\tilde{\varphi} : \pi^{-1}(U) \rightarrow V \times \mathbb{R}^m, \quad \xi_p \mapsto \left( \varphi(p), \begin{pmatrix} \xi_p(\varphi_1) \\ \vdots \\ \xi_p(\varphi_m) \end{pmatrix} \right).$$

Now we declare a subset  $W \subseteq TM$  to be open, if and only if for all charts  $\varphi$  the set  $\tilde{\varphi}(W \cap \pi^{-1}(U)) \subset V \times \mathbb{R}^m$  is open. We can now extend and give meaning to the notation of the local basis vectors  $\partial_i(p) \in T_p M$ : For any chart  $\varphi : U \rightarrow V$  we have the local sections

$$\partial_i : U \rightarrow TM, \quad p \mapsto \partial_i(p).$$

Furthermore,  $TM$  is a  $2m$ -dimensional manifold. Using a smooth atlas on  $M$  yields a smooth atlas on  $TM$  making it a smooth manifold. Notice that the family  $(\partial_i)_i$  is a basis of the  $C(U)$ -module  $\Gamma(U)$ . This follows by expressing any section in local coordinates via  $\tilde{\varphi}$ .

Notice that  $TM$  as a bundle satisfies more than just the bundle definition: Taking two different charts induces two different local trivialisations. Transitioning between them is not just fibrewise linear, but also smooth. This is captured in the definition:

Henceforth we suppress the differentiable structure from the notation.

**Definition 2.1.19** (The Derivative). Let  $M$  and  $M'$  be smooth manifolds. We call a continuous function  $f : M \rightarrow M'$  **smooth**, if for all  $\varphi \in \mathfrak{A} \in \mathcal{C}$  and  $\varphi' \in \mathfrak{A}' \in \mathcal{C}'$  the maps

$$\varphi' \circ f \circ \varphi^{-1} : V \rightarrow V'$$

are smooth. A given smooth function induces a smooth map between the tangent bundles as manifolds as follows:

$$df : TM \rightarrow TM', df_p(\xi_p)(g_p) = \xi((g \circ f)_p)$$

Here,  $\xi_p \in T_p M$ ,  $g_p \in \mathcal{E}_p(M')$ . The function  $df$  is called the **derivative** or **pushforward** of  $f$ .

**Definition 2.1.20** (Smooth Vector Bundles). A **smooth vector bundle of rank  $r$**  is a surjection  $\pi : E \rightarrow M$  where the codomain (or base space) is a smooth manifold together with a family of continuous functions (or local trivialisations)

$$\Phi_i : \pi^{-1}(U_i) \rightarrow U_i \times \mathbb{R}^r$$

such that  $\bigcup_i U_i = M$  and for all overlapping  $U_i \cap U_j \neq \emptyset$ , the transition functions

$$\Phi_j \circ \Phi_i^{-1} : \pi^{-1}(U_i \cap U_j) \rightarrow (U_i \cap U_j) \times \mathbb{R}^r$$

are of the form

$$\Phi_j \circ \Phi_i^{-1}(p, v) = (p, g_{i,j}(v)), \quad \text{where} \quad g_{i,j} : (U_i \cap U_j) \rightarrow \text{GL}(r, \mathbb{R})$$

is smooth.

**Corollary 2.1.21.** A smooth vector bundle is a smooth manifold and also a vector bundle. The latter is ensured by the codomain of  $g_{i,j}$  which is also called the **structure group**. We can enforce structure on the bundle by reducing the structure group. As a preview: reducing it to  $\text{GL}^+(k, \mathbb{R})$  the linear maps with positive determinant means that we can induce the concept of orientations. Reducing it to  $\text{O}(k)$  lets us equip it with a Riemannian metric.

def:vb-frames  
FromTrivialis  
ations

**Definition 2.1.22.** Let  $\pi : E \rightarrow M$  be a smooth vector bundle of rank  $r$  and let

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r$$

be a local trivialisation. Then a section is called smooth if it is a smooth map of manifolds. A family of smooth local sections  $(s_1, \dots, s_r)$  with  $s_i \in \Gamma(U)$  is called a **local frame** if they form a basis of the  $C(U)$ -module  $\Gamma(U)$ . Such a family is obtained from  $\Phi$  by defining  $s_i(p) := \Phi^{-1}(p, e_i)$ . Similar, from a local frame we can obtain a local trivialisations by defining:

$$\Phi : \pi^{-1}(U) \rightarrow U \times \mathbb{R}^r, \quad v = \sum_{i=1}^r \lambda_i s_i(\pi(v)) \mapsto (\pi(v), \lambda_1, \dots, \lambda_r).$$

**Example 2.1.23.** The local trivialisations  $\partial_i$  of the tangent bundle of a smooth manifold  $M$  form a local frame and make the tangent bundle into a smooth vector bundle.

**Definition 2.1.24** (Compact-open topology). Let  $X, Y$  be topological spaces. Then the set of continuous maps  $C(X, Y)$  can be equipped with the topology that has

$$\{V(K, U) | K \subseteq X \text{ compact}, U \subseteq Y \text{ open}\} \text{ where } V(K, U) := \{f \in C(X, Y) | f(K) \subseteq U\}$$

as a subbasis. This coincides with the topology of  $\text{Hom}(V, W)$  as a subset of  $\mathbb{C}^n$  after identification with the corresponding matrix.

def:vb-homomorphismBundle

**Definition 2.1.25** (Homomorphism Bundle). Let  $W, V$  be vector spaces and  $E = V \times X$ ,  $F = W \times X$  the product bundles over  $X$ . Now any bundle homomorphism  $\varphi : E \rightarrow F$  determines a map  $\Phi : X \rightarrow \text{hom}(V, W)$  by sending  $x \mapsto \varphi_x$ . With respect to the compact-open topology, this  $\Phi$  is continuous. Furthermore, a continuous map  $\Phi : X \rightarrow \text{hom}(V, W)$  determines a bundle homomorphism  $\varphi : E \rightarrow F$ .

*Proof.* Let  $\{e_i\}$  and  $\{f_i\}$  be bases of  $V$  and  $W$ . Then each  $\Phi(x)$  is a matrix  $\Phi_{i,j}(x)$ , such that:

$$\Phi(x)e_i = \sum_j \Phi_{i,j}(x)f_j.$$

Now  $\Phi$  is continuous if and only if  $\Phi_{i,j}$  is continuous. Since  $\varphi(x, v) = (x, \Phi(x)v)$  this is equivalent to  $\varphi$  being continuous.  $\square$

thm:vb-isomorphicVectorBundles

**Lemma 2.1.26** (Isomorphic Vector Bundles). *Let  $E$  and  $F$  be vector bundles and  $\varphi : E \rightarrow F$  be a homomorphism. Then  $\varphi$  is an isomorphism if and only if all  $\varphi_x$  are isomorphisms.*

*Proof.* To show that it is an isomorphism we need a continuous inverse, equivalently, we need to show that  $\varphi$  is a homeomorphism. First of all we can deduce, that  $\varphi$  is bijective: Assume it wasn't injective, then there would be a fibre in which  $\varphi_x$  is not injective. Assume that it wasn't surjective. Then there would be a  $w \in F$  that has no preimage which means that in the fiber containing  $w$  there is no isomorphism. Thus, there is a unique inverse and it remains to verify that it is continuous, which is a local condition. Therefore we can assume that  $E$  and  $F$  are product bundles. Then we have a continuous map  $\Phi : X \rightarrow \text{Hom}(V, W)$ . Furthermore, since all  $\varphi_x$  are isomorphisms, we can restrict the image to  $\text{Iso}(V, W)$  which is an open subset of  $\text{Hom}(V, W)$ . Now we can define the map  $x \mapsto \Phi(x)^{-1}$  which is continuous since taking inverses is continuous. By the corollary above, we obtain a continuous map  $\psi : F \rightarrow E$  by sending  $(x, w) \mapsto (x, \Phi(x)^{-1}(w))$ , which is the inverse.  $\square$

**Example 2.1.27.** Let  $M$  be a smooth manifold of dimension  $m$ ,  $\varphi : U \rightarrow V$  a chart and  $\tilde{\varphi} : TU \rightarrow V \times \mathbb{R}^m$  be the corresponding bundle chart. Then it follows from the lemma, that  $TU$  is isomorphic to  $V \times \mathbb{R}^m$ . This is verified by observing that  $\partial_i(p)$  is a basis of  $T_p U$  for all  $p \in U$ . In this case  $\tilde{\varphi}(\partial_i) = e_i$  giving us isomorphisms in each fiber.

Now we want to construct new bundles from old ones. There are many operations to construct new vector spaces such as the tensor product, direct sum, Hom-Spaces and many more. Instead of individually proving that those operations applied fiberwise yield new bundles, we subsume these operations within the following definition:

**Definition 2.1.28** (Continuous Functor). Let  $T$  be a covariant endofunctor in the category of finite-dimensional vector spaces. We call  $T$  a **continuous functor**, if for all  $V, W$  the map  $T : \text{hom}(V, W) \rightarrow \text{hom}(T(V), T(W))$  is continuous. All vector spaces are given the topology induced by their isomorphism to real Euclidean space. If  $E$  is a vector bundle we define the set:

$$T(E) := \bigcup_{x \in X} T(E_x).$$

Now we want to equip this with a topology such that  $T(E)$  is a bundle over the same base space as  $E$ .

def:vb-constructingNewBundles

**Definition 2.1.29** (Constructing new bundles). We start by assuming that  $E = X \times V$ . In that case we define the topology on  $T(E)$  as the product topology  $X \times T(V)$ . Now suppose that  $F = X \times W$  is another product bundle and  $f : E \rightarrow F$  is a homomorphism. Let  $\Phi : X \rightarrow \text{hom}(V, W)$  be the corresponding map. Then  $T\Phi : X \rightarrow \text{hom}(T(V), T(W))$  is continuous by assumption. Thereby,  $T(\varphi) : X \times T(V) \rightarrow X \times T(W)$  is continuous. If  $\varphi$  is an isomorphism, then  $T(\varphi)$  is also an isomorphism. That follows because it is continuous and an isomorphism fibrewise.

Now assume that  $E$  is trivial. Then we can choose an isomorphism  $\alpha : E \rightarrow X \times V$  and induce a topology on  $T(E)$  by pulling it back via  $T(\alpha) : T(E) \rightarrow X \times T(V)$ . This topology does not depend on  $\alpha$  by the above. For general vector bundles we do this construction locally.

If  $f : X \rightarrow Y$  is continuous, there is a natural isomorphism

$$f^*T(E) \cong Tf^*(E)$$

This construction is carried out in detail by Atiyah in [Ati89, Chapter 1.2] and can be generalised for continuous functors with several variables both co- and contravariant. Hence this procedure gives us for vector bundles  $E$  and  $F$  over the same base space the new bundles:

- (i)  $E \oplus F$ , their direct sum
- (ii)  $E \otimes F$ , their tensor product
- (iii)  $\text{Hom}(E, F)$
- (iv)  $E^*$ , the dual bundle of  $E$
- (v)  $\Lambda^i(E)$  the  $i$ -th exterior power

and by the natural isomorphism we have natural isomorphisms:

- (i)  $E \oplus F \cong F \oplus E$
- (ii)  $E \otimes F \cong F \otimes E$
- (iii)  $E \otimes (F \oplus F') \cong E \otimes F \oplus E \otimes F'$
- (iv)  $\text{Hom}(E, F) \cong E^* \otimes F$

cor:vb-sectionsIntoHom

**Corollary 2.1.30.** Let  $\pi_E : E \rightarrow X$  and  $\pi_F : F \rightarrow X$  be two vector bundles over  $X$ . Then, the space  $\Gamma(\text{Hom}(E, F))$  can be identified with the space of bundle morphisms  $\text{hom}(E, F)$  since for any section  $s : X \rightarrow \text{Hom}(E, F)$  we can define the bundle homomorphism  $\tilde{s} : E \rightarrow F$  by sending  $v \mapsto s_{\pi_E(v)}(v)$ . This is a homomorphism since  $\pi_F(\tilde{s}(v)) = \pi_E(v)$ .

**Example 2.1.31** (Cotangent bundle). Let  $M$  be a smooth manifold of dimension  $m$ . Then we can define the **cotangent bundle**  $T^*M$  to be the dual of the tangent bundle. A local frame of said bundle is given by the fiberwise dual of the frame  $(\partial_i)$ . Sections into the tangent bundle are called **vector fields** and sections into the cotangent bundle are called **one-forms**.

def:vb-linear-identification

**Definition 2.1.32.** Let  $V$  be a linear manifold (i.e. a finite-dimensional vector space). Then for any  $v \in V$  we have the canonical maps

$$R_v : V \rightarrow T_v V$$

$$x \mapsto \left( f_v \mapsto \frac{\partial}{\partial t} \Big|_{t=0} f(v + tx) \right)$$

Let  $L : V \rightarrow V$  be a linear map viewed as a smooth map between manifolds. Then we have the identity:

$$dL_v \circ R_v = R_{L(v)} \circ L.$$

This is proved by a short calculation:

$$\left( dL_v(R_v(x)) \right) (f_{L(v)}) = R_v(x) (f \circ L) = \frac{\partial}{\partial t} \Big|_{t=0} f \circ L(v + tx) = \underbrace{\frac{\partial}{\partial t} \Big|_{t=0} f(L(v) + tL(x))}_{=R_{L(v)} \circ L(x)}$$

Now let  $v, v' \in V$ . Then we have the translation

$$tr_{v,v'} : V \rightarrow V; \quad x \mapsto (x + (v' - v))$$

This allows us to relate  $R_v$  and  $R_{v'}$  as follows:

$$\left( dtr_{v,v'}|_v (R_v(x)) \right) (f_{v'}) = \frac{\partial}{\partial t} \Big|_{t=0} (f \circ tr_{v,v'}(v + tx)) = \frac{\partial}{\partial t} \Big|_{t=0} (f(v' + tx)) = R_{v'}(x)(f_{v'}).$$

Hence:

$$R_{v'} = dtr_{v,v'} \circ R_v.$$

The **linear identification** is the map  $Q$  given on the whole tangent bundle:

$$Q : TV \rightarrow V; \quad \xi_p \mapsto R_p^{-1}(\xi_p),$$

and satisfies for any linear  $L : v \rightarrow V$

$$Q \circ dL = L \circ Q.$$

cor:vb-vector  
FieldsTakeFun  
ctions

**Corollary 2.1.33.** Let  $f : M \rightarrow \mathbb{R}$  be smooth. Then we can interpret  $df$  as a one-form. To be precise, let  $Q : T\mathbb{R} \rightarrow \mathbb{R}$  denote the linear identification with respect to the identity on  $\mathbb{R}$ . For  $v \in \Gamma(TM)$  we have

$$(Q \circ df)(v) = v(f)$$

Here on the right side,  $f$  denotes a map  $p \mapsto f_p$  such that  $v(f)(p) := v_p(f_p)$  is well defined. We retain this notation so that vector fields can take functions as an input.

*Proof.* Let  $\varphi : U \rightarrow V$  be a chart of  $M$ . We prove this by showing it for the local sections  $\partial_i$  and the general case follows by  $C(U)$ -linearity, since those sections form a local frame. Now let  $g_p \in \mathcal{E}_p(\mathbb{R})$  and  $\varphi(p) = x_0$ . Then

$$df_p(\partial_i(p))(g_p) = \partial_i((g \circ f)_p) = \frac{\partial}{\partial x_i} \Big|_{x_0} (g \circ f \circ \varphi^{-1}) = \frac{\partial}{\partial x_{f_p(p)}} (g_p) \cdot \partial_i(p)(f_p)$$

Hence, applying  $Q$  to the calculation yields  $(Q \circ df)(v) = v(f)$ .  $\square$

**Definition 2.1.34** (A Metric). Let  $M$  be a smooth manifold. A section  $g \in \Gamma(T^*M \otimes T^*M)$  into the tensor product of the dual tangent bundle with itself is a **Riemannian metric** if the following are satisfied for all  $p \in M$ :

1.  $g$  is **non-degenerate**, meaning if for a  $v_p \in T_pM$  we have  $g_p(v_p, w_p) = 0$  for all  $w_p \in T_pM$  then  $v_p = 0$ .
2.  $g$  is **symmetric**, meaning  $g_p(v_p, w_p) = g_p(w_p, v_p)$  for all  $v_p, w_p \in T_pM$ .
3.  $g$  is **positive definite**, meaning that  $g_p(v_p, v_p) \geq 0$  for all  $v_p$  with equality only for  $v_p = 0$ .

We call a tuple  $(M, g)$  a **Riemannian manifold**.

**Remark 2.1.35.** Equipping a smooth manifold  $M$  of dimension  $m$  with a Riemannian metric is equivalent to reducing the structure group of the tangent bundle from  $GL(m)$  to  $O(m)$ . To be precise: A Riemannian metric  $g$  lets us canonically choose an atlas with structure group  $O(m)$  and an atlas with such a structure group lets us canonically choose a Riemannian metric.



*Proof.* We start by assuming that we have an atlas with structure group  $O(m)$ . In Definition 2.1.22 we saw that a covering of local trivializations is equivalent to a covering of smooth frames. Now in a single trivialization

$$\Phi : \pi^{-1}(U) \cong TU \rightarrow U \times \mathbb{R}^m$$

we call the projection to the  $\mathbb{R}^m$ -part (the second factor)  $\pi_2$  and define

$$\begin{aligned} g : U &\rightarrow T^*U \otimes T^*U \\ p &\mapsto g_p^\Phi : T_pU \times T_pU \rightarrow \mathbb{R} \\ &\quad (\nu_p, \xi_p) \mapsto g_p^\Phi(\nu_p, \xi_p) := \pi_2(\Phi(\nu_p)) \cdot \pi_2(\Phi(\xi_p)), \end{aligned}$$

where  $\cdot$  denotes the standard scalar product. This gives us locally a Riemannian metric. If we do this for all trivializations we have to make sure, that the result is well defined. So assume that two trivializations  $\Phi$  and  $\Psi$  overlap in  $V$ . The structure group is  $O(m)$ ,  $\Phi = R\Psi$  for some function

$$R : V \rightarrow O(m).$$

Let  $p$  be in said overlapping and  $(\nu_p, \xi_p)$  be from  $T_pM \times T_pM$ . Then since  $R$  is orthogonal we can calculate the independence from the trivialization:

$$g_p^\Phi(\nu_p, \xi_p) = \pi_2(R\Psi(\nu_p)) \cdot \pi_2(R\Psi(\xi_p)) = \pi_2(\Psi(\nu_p)) \cdot \pi_2(\Psi(\xi_p)) = g_p^\Psi(\nu_p, \xi_p).$$

To get an orthonormal atlas from a smooth one given a Riemannian metric, we start by building a covering of local frames from the trivialisations. We then apply Gram-Schmidt to the local frames to get orthonormal frames. Now a covering of orthonormal frames gives a covering of trivializations, where the structure group consists of basis changes between orthonormal bases and hence is  $O(m)$ .  $\square$

**Definition 2.1.36** (The Gradient). Assume that  $(M, g)$  is a smooth manifold together with a Riemannian metric. Let  $f : M \rightarrow \mathbb{R}$  be a smooth function and  $Q : T\mathbb{R} \rightarrow \mathbb{R}$  the linear identification. We define its **gradient** to be a vector field  $\text{grad}(f) \in \Gamma(TM)$  such that for any vector field  $V$ :

$$Q \circ df(V) = g(\text{grad}(f), V).$$

Notice how the non-degeneracy of  $g$  ensures existence and uniqueness.

**Definition 2.1.37.** Let  $(M, g)$  be a Riemannian manifold and  $\varphi : U \rightarrow V$  be a chart. We define the smooth functions  $g_{ij} : U \rightarrow \mathbb{R}$  to be:

$$g_{ij}(p) = g_p(\partial_i(p), \partial_j(p)).$$

Then if two vector fields  $v, w$  over  $U$  are given with  $v = \sum v_i \partial_i$  and  $w = \sum w_i \partial_i$  we can calculate the metric locally:

$$g(v, w) = \sum_{i,j} g_{ij} v_i w_j : U \rightarrow \mathbb{R}$$

def:vb-localD  
escriptionOfG  
radientGij

Furthermore, we can invert the matrices  $(g_{ij}(p))_{ij}$  for all  $p$  (which is a smooth procedure as can be seen by the construction via Cramer's rule) to get smooth functions

$$g^{ij} : U \rightarrow \mathbb{R}$$

$$p \mapsto g^{ij}(p) = \left( (g_{kl}(p))_{kl}^{-1} \right)_{ij}$$

**Remark 2.1.38.** Due to the rare appearance of big summations with vectors and co-vectors in this work, we omit the Einstein summation convention and assign all coordinate entries lower indices.

lem:vb-localD  
escriptionOfG  
radient

**Lemma 2.1.39** (Local Description of the Gradient). *Let  $(M, g)$  be a smooth Riemannian manifold of dimension  $m$  and  $\varphi : U \rightarrow V$  be a chart. Then the gradient has the local form:*

$$\text{grad}(f) = \sum_{i,j} g^{ij} (\partial_i(f)) \cdot \partial_j.$$

*Proof.* Assume that  $v, w \in \Gamma(TU)$  such that  $v = \sum_i v_i \partial_i$  and  $w = \sum_i w_i \partial_i$  (Here  $v_i$  and  $w_i$  are smooth functions  $U \rightarrow \mathbb{R}$ ). Then  $g(v, w) = \sum_{i,j} g_{ij} v_i w_j$ . Suppose that  $\text{grad}(f) = \sum_j G^j \partial_j$  and  $Q$  is the linear identification  $T\mathbb{R} \rightarrow \mathbb{R}$  in each tangent space. Then by definition of the gradient and corollary 2.1.33 we have:

$$\partial_j(f) = Q \circ df(\partial_j) = g(\text{grad}(f), \partial_j) = g\left(\sum_i G^i \partial_i, \partial_j\right) = \sum_i g_{ij} G^i$$

Reading this as a pointwise matrix product:

$$(G^1, \dots, G^m)(g_{ij}) = (\partial_1(f), \dots, \partial_m(f))$$

which gives  $(G^1, \dots, G^m) = (\partial_1(f), \dots, \partial_m(f)) (g^{ij})$ .

But then we can conclude that  $G^j = \sum_i g^{ij} \partial_i(f)$ .

□

**Definition 2.1.40** (A Dynamical System). Let  $M$  be a smooth manifold. A **dynamical system on  $M$**  (sometimes also called a **flow on  $M$** ) is given by a smooth map  $\varphi : \Omega \rightarrow M$  such that:

1.  $\{0\} \times M \subseteq \Omega \subseteq \mathbb{R} \times M$  open, and for any  $p \in M$  the set  $I(p) := \pi_1(\Omega \cap \mathbb{R} \times \{p\})$  is connected. ( $\pi_1$  is the projection onto the first factor)
2. For all  $p \in M$  and  $t \in I(p)$  we have that  $s \in I(\varphi(t, p))$  if and only if  $s + t \in I(p)$ . Then we have:

$$\varphi(s, (\varphi(t, p))) = \varphi(s + t, p)$$

3. For all  $p$  we require  $\varphi(0, p) = p$ .

We will write  $\varphi_t(p) := \varphi(t, p)$  to keep the notation manageable. If for all  $p$   $I(p) = \mathbb{R}$  we call the system **global**.

**Corollary 2.1.41.** For a global dynamical system  $\varphi$  on a smooth manifold  $M$  the map

$$\rho : \mathbb{R} \rightarrow \text{Diff}(M), \quad t \mapsto \varphi_t$$

is a homomorphism. To verify well-definedness notice how for all  $t \in \mathbb{R}$ ,  $\varphi_{-t}$  is an inverse of  $\varphi_t$ . The map  $\rho$  is called a **one-parameter group of diffeomorphisms**.

**Definition 2.1.42** (Vector Field of a Dynamical System). Let  $\varphi : \Omega \rightarrow M$  be a dynamical system on a smooth manifold  $M$ . For a chart  $\psi : U \rightarrow V$  of  $M$  we have the chart  $\text{id} \times \psi$  of  $\Omega$ . For this chart we denote the associated local frame by  $(\partial_t, \partial_1, \dots, \partial_m)$ . Using this we define the mapping

$$\begin{aligned} M &\rightarrow TM \\ p &\mapsto X(p) := d\varphi(\partial_t(0, p)) \end{aligned}$$

and call it the **associated vector field**. This is in fact a smooth vector field.

**Definition 2.1.43** (Dynamical System of a Vector Field). Let  $X \in \Gamma(TM)$  be a global vector field on the smooth manifold  $M$ . Then we call a dynamical system  $\varphi$  on  $M$  the **associated dynamical system** to  $X$ , if for all  $t \in \mathbb{R}$  and  $x \in M$

$$d\varphi(\partial_t(t, x)) = X(\varphi_t(x)).$$

**Remark 2.1.44.** Regarding the left-hand side of the definition  $d\varphi(\partial_t(t, x))$  we could also differentiate the map

$$t \mapsto \varphi_t(x),$$

for fixed  $x$  at  $t$  and evaluate this at the constant vector field  $t \mapsto R_t(1) \in T_t\mathbb{R}$  from definition 2.1.32.

These two constructions are mutually inverse, giving the following theorem:

**Theorem 2.1.45.** *Let  $M$  be a smooth manifold. Then there is a one-to-one correspondence between global dynamical systems and vector fields given by the constructions above.*

## 3 Complex K Theory

### 3.1 Mathematical Goals

**Theorem 3.1.1** (The Theorems and Mathematical Structures). • *Definition of The K- Functor*

- *Cohomology structures*
  - *Additivity (pointed sum)*
  - *Long exact sequences*
  - *Excision*

### 3.2 Classification of Vector Bundles

The first goal is to classify all vector bundles over sufficiently nice spaces. The latter refers to spaces admitting partitions of unity for which Tietze's extension theorem holds. For simplicity we just consider compact Hausdorff spaces. Along the way we will discover that the set of isomorphism classes of vector bundles is a homotopy invariance.

Before we go back to the analytical reals, we will further develop the theory of vector bundles:

**Definition 3.2.1.** For a topological space  $X$  we define  $\text{Vect}_n$  to be the set of isomorphism classes of complex vector bundles of rank  $n$  and  $\text{Vect}(X)$  to be the set of isomorphism classes of complex vector bundles of any rank. By declaring  $\text{Vect}(f) = f^*$  to be the pullback we get a contravariant functor  $\text{Vect}$  from **Top** to **Set**. We can equip the set  $\text{Vect}(X)$  with the direct sum and tensor product operation. It is then easy to see that these operations descend to isomorphism classes and hence we get an abelian semi ring  $(\text{Vect}(X), \oplus, \otimes)$  (abelian semigroup with associative and distributive multiplication). In fact we even get a contravariant functor from **Top** to the category of abelian semirings.

lem:kTheo-sectionExtension

**Lemma 3.2.2.** *Let  $X$  be a compact Hausdorff space,  $Y \subseteq X$  a closed subspace and  $E$  a vector bundle over  $X$ . Then any section  $s: Y \rightarrow E|_Y$  can be extended to a global section of  $X$ .*

*Proof.* For a detailed proof see Chapter 1.4 of [Ati89]. The general argument however is easy to follow: By Tietze's extension theorem we can extend the section to around every  $x \in X$ . By compactness we can pick finitely many of those extensions. Now using a partition of unity with respect to their domains we can build the sum of all the extensions to get a well defined global one.  $\square$

Atiyah1989k

cor:kTheo-isomorphismExtension

**Corollary 3.2.3.** Let  $Y$  be a closed subspace of a compact Hausdorff space  $X$ , and let  $E, F$  be bundles over  $X$ . If  $f: E|_Y \rightarrow F|_Y$  is an isomorphism, then there is an open neighbourhood  $U \supset Y$  and an extension  $f: E|_U \rightarrow F|_U$  which is an isomorphism.

*Proof.* By corollary 2.1.30 we can view  $f$  as a section into the bundle  $\text{Hom}(E|_Y, F|_Y)$ , which can be extended to a global one in  $\text{Hom}(E, F)$ . Let  $U$  be the open set for which the section maps to isomorphisms. Since  $U$  contains  $Y$  this yields the require extension.  $\square$

thm:kTheo-hom  
otopyInvarian  
ce

**Theorem 3.2.4** (Homotopy Invariance of the Pullback). *Let  $X, Y$  be compact Hausdorff spaces,  $I = [0, 1]$ , and*

$$f : Y \times I \rightarrow X$$

*be a homotopy. For a fixed  $t \in I$ , we define the function  $f_t : Y \rightarrow X$ ,  $f_t(y) = f(t, y)$ . Let  $\pi : E \rightarrow X$  be a vector bundle. Then*

$$f_0^*(E) \cong f_1^*(E).$$

*Proof.* Consider the following two vector bundles over  $Y \times I$ :

$$f^*E \quad \text{and} \quad \pi_1^* f_t^* E,$$

where  $\pi_1 : Y \times I \rightarrow Y$  denotes the projection and  $t \in I$ . Since the restriction is by definition the pullback along the inclusion  $i : Y \times \{t\} \rightarrow Y \times I$ . The equality of the two maps:

$$f \circ i = f_t \circ \pi_1 \circ i : Y \times \{t\} \rightarrow X$$

together with the functoriality gives us an isomorphism

$$\Phi : f^*E|_{Y \times \{t\}} \rightarrow \pi_1^* f_t^* E|_{Y \times \{t\}}.$$

By Corollary 3.2.3 we can extend the isomorphism for some open neighbourhood of  $Y \times \{t\}$  and by the product topology we can deduce that there exists a  $\delta_t$  such that  $U$  contains the thickened line  $Y \times (t - \delta_t, t + \delta_t)$ . The isomorphism class of  $f^*E|_{Y \times \{t\}}$  as a bundle over  $Y$  is locally constant in  $t$ : Let  $0 < \varepsilon < \delta$  and  $r \in I$ . Then define the inclusions

$$\begin{array}{ccc} i_t : Y & \rightarrow & Y \times \{t\} \subseteq Y \times I \\ y & \mapsto & (y, t) \end{array} \quad \text{and} \quad \begin{array}{ccc} i_{t+\varepsilon} : Y & \rightarrow & Y \times (t - \delta_t, t + \delta_t) \\ y & \mapsto & (y, t + \varepsilon) \end{array}$$

Then we have the identities  $f_t = f \circ i_t$  and  $\pi_1 \circ i_{t+\varepsilon} = \text{id}_Y$  and we can calculate:

$$f_{t+\varepsilon}^* E = i_{t+\varepsilon}^* f^* E = i_{t+\varepsilon}^* \pi_1^* f_t^* E = (f_t \circ \pi_1 \circ i_{t+\varepsilon})^* E = f_t^* E.$$

Finally, since  $I$  is connected, any locally constant map from  $I$  is globally constant and therefore  $f_0^* E \cong f_1^* E$ .  $\square$

lem:kTheo-con  
tractibleBund  
lesAreTivial

**Corollary 3.2.5.** 1. If  $f : X \rightarrow Y$  is a homotopy equivalence,  $f^* : (Y) \rightarrow (X)$  is a semiring isomorphism.

2. If  $X$  is contractible, every bundle over  $X$  is trivial and  $(X) \cong \mathbb{N}_0$

def:kTheo-tri  
vialization

**Definition 3.2.6** (Trivialization of a Vector Bundle). Let  $Y$  be a closed subspace of  $X$ ,  $E$  a vector bundle over  $X$  and  $\alpha : E|_Y \rightarrow Y \times V$  be an isomorphism. Then we call  $\alpha$  **trivialisation of  $E$  over  $Y$** .

lem:kTheo-tri  
vialisationDe  
finesABundle

**Lemma 3.2.7.** *A trivialization  $\alpha$  of a bundle  $E$  over  $Y \subseteq X$  defines a bundle  $E/\alpha$  over  $X/Y$ . The isomorphism class of  $E/\alpha$  depends only on the homotopy class of  $\alpha$ .*

*Proof.* We define the bundle  $E/\alpha$  as follows: Let  $\pi_2 : Y \times V \rightarrow V$  denote the projection and define an equivalence relation on  $E|_Y$  by

$$e \sim e' \Leftrightarrow \pi_2 \alpha(e) = \pi_2 \alpha(e').$$

We extend this relation by the identity on  $E|_{X \setminus Y}$  and we define  $E/\alpha$  to denote the quotient space of  $E$  defined by that relation. To see why this is a bundle we need to check the local triviality only around the base point  $Y/Y$  of  $X/Y$ . This follows from the fact that by Corollary 3.2.3 we can extend  $\alpha$  to an isomorphism  $\tilde{\alpha} : E|_U \rightarrow U \times V$ . Then  $\tilde{\alpha}$  induces an isomorphism

$$E/\alpha|_{U/Y} \cong (U/Y) \times V$$

proving the local triviality. Now for the homotopy invariance of said bundle suppose that  $\alpha_0$  and  $\alpha_1$  are homotopic trivializations of  $E$  over  $Y$ . This means that we have a trivialization  $\beta$  of  $E \times I$  over  $Y \times I \subset X \times I$  that induces the  $\alpha_i$  at the endpoints. Let  $f : (X/Y) \times I \rightarrow (X \times I)/(Y \times I)$  be the natural map. Then  $f^*((E \times I)/\beta)$  is a bundle on  $(X/Y) \times I$  whose restriction to  $(X/Y) \times \{i\}$  is  $E/\alpha_i$  ( $i = 0, 1$ ). Hence by the homotopy invariance of the pullback (Theorem 3.2.4) we have

$$E/\alpha_0 \cong E/\alpha_1.$$

□

lem:kTheo-col  
lapsingOfCont  
ractableSubsp  
ace

**Lemma 3.2.8.** *Let  $Y \subseteq X$  be a closed contractible subspace of a compact Hausdorff space  $X$ . Then*

$$f : X \rightarrow X/Y \quad \text{induces a semiring isomorphism} \quad f^* : \text{Vect}(X/Y) \rightarrow \text{Vect}(X).$$

*Proof.* Since  $Y$  is contractible,  $E|_Y$  is trivial and thereby a trivialization  $\alpha : E|_Y \rightarrow Y \times V$  exists. Two such trivializations only differ by the automorphism of  $Y \times V$  which in turn corresponds to a map  $Y \rightarrow \text{GL}(V) \cong \text{GL}_{\text{rank}(E)}(\mathbb{C})$ . Now since  $Y$  is contractible those two maps factor (up to homotopy) over any one point  $y \in Y$  and are thereby homotopically equivalent to the constant map. Finally since  $\text{GL}_{\text{rank}(E)}(\mathbb{C})$  is connected all constant maps are homotopically equivalent. Thereby  $\alpha$  is unique up to homotopy. By Lemma 3.2.7 the isomorphism class of  $E/\alpha$  only depends on the homotopy class of  $\alpha$  we have a well defined map

$$\text{Vect}(X) \rightarrow \text{Vect}(X/Y),$$

which in turn is an inverse of  $f^*$  concluding the proof. □

def:kTheo-gluingAndClutching

**Definition 3.2.9** (Gluing and Clutching of Vector Bundles). Let

$$X = X_1 \cup X_2, \quad A = X_1 \cap X_2,$$

and all be compact Hausdorff spaces. Assume that  $E_i$  is a vector bundle over  $X_i$  and  $\varphi : E_1|_A \rightarrow E_2|_A$  is an isomorphism. Then we define a vector bundle  $E_1 \cap_\varphi E_2$  on  $X$  as follows: We topologize the space  $E_1 \cup_\varphi E_2$  as the quotient of  $E_1 + E_2 / \sim$ , where  $\sim$  is the identification of points with their image under  $\varphi$ . Identifying  $X$  with  $X_1 + X_2$  in the quotient space we get the natural projection  $p : E_1 \cap_\varphi E_2 \rightarrow X$  and the fibers have a natural vector space structure. It remains to show that this is locally trivial. This is obvious outside of  $A$  so let  $a \in A$  and  $V_1$  be a closed neighbourhood of  $a$  in  $X_1$  such that  $E_1|_{V_1}$  is trivial giving us the isomorphisms

$$\begin{aligned} \theta_1 : E_1|_{V_1} &\rightarrow V_1 \times \mathbb{C}^n \\ \theta_1^A : E_1|_{V_1 \cap A} &\rightarrow (V_1 \cap A) \times \mathbb{C}^n \end{aligned}$$

Finally, define

$$\theta_2^A : E_2|_{V_1 \cap A} \rightarrow (V_1 \cap A) \times \mathbb{C}^n$$

by composition  $\theta_1^A \circ \varphi^{-1}$ . Let  $V_2$  be a neighborhood of  $a$  in  $X_2$  such that

$$\theta_2 : E_2|_{V_2} \rightarrow V_2 \times \mathbb{C}^n$$

is an extension of  $\theta_2^A$ . This extension exists again by the extending the section into the homomorphism bundle corresponding to the isomorphism  $\theta_2^A$ . Finally  $\theta_1$  and  $\theta_2$  defines a well defined isomorphism

$$\theta_1 \cap_\varphi \theta_2 : E_1 \cap_\varphi E_2 \rightarrow (V_1 \cap V_2) \times \mathbb{C}^n.$$

This proves the local triviality.

We call such a triple  $(E_1, E_2, \varphi)$  a **clutching datum** and  $\varphi$  the **transition function**.

cor:kTheo-elementaryPropertiesGluingAndClutching

**Corollary 3.2.10** (Elementary Properties of the Gluing and Clutching Construction). Elementary properties of the gluing and clutching construction are:

1. If  $E$  is a bundle over  $X$ , then the identity defines an isomorphism and

$$E_1 \cup_{\text{id}_A} E_2 \cong E.$$

2. If  $\beta_i : E_i \rightarrow E'_i$  are isomorphisms on  $X_i$  and  $\varphi' \beta_1 = \beta_2 \varphi$ , then

$$E_1 \cup_\varphi E_2 \cong E'_1 \cup_{\varphi'} E'_2.$$

3. If  $(E_i, \varphi)$  and  $(E'_i, \varphi')$  are two "clutching data" on  $X_i$ , then

$$(E_1 \cup_\varphi E_2) \oplus (E'_1 \cup_{\varphi'} E'_2) \cong E_1 \oplus E'_1 \bigcup_{\varphi \oplus \varphi'} E_2 \oplus E'_2,$$

$$(E_1 \cup_\varphi E_2) \otimes (E'_1 \cup_{\varphi'} E'_2) \cong E_1 \otimes E'_1 \bigcup_{\varphi \otimes \varphi'} E_2 \otimes E'_2,$$

$$(E_1 \cup_\varphi E_2)^* \cong E_1^* \bigcup_{(\varphi^*)^{-1}} E_2^*.$$

$(-)^*$  denotes the dual here.

lem:kTheo-hom  
otopyInvariant  
ceClutchingFu  
nction

**Lemma 3.2.11.** *The isomorphism class of  $E_1 \cup_\varphi E_2$  depends only on the homotopy class of the isomorphism  $\varphi : E_1|_A \rightarrow E_2|_A$ .*

*Proof.* A homotopy of isomorphisms is a map

$$\Phi : E_1|_A \times I \rightarrow E_2|_A ,$$

such that for all  $t$  the map

$$\Phi_t : E_1|_A \rightarrow E_2|_A \quad \Phi_t(\xi) := \Phi(\xi, t)$$

is a bundle isomorphism. We can encode this  $\Phi$  into a bundle isomorphism as follows: Let  $\pi : X \times I \rightarrow X$  be the projection. Then elements  $\xi \in \pi^* E_1$  are by definition 2.1.13 of the form  $\xi_{x,t} = ((x, t), \xi_x) \in (\pi^* E_1)_{x,t}$ . With this notation define the bundle isomorphism:

$$\Psi : \pi^*(E_1)|_{A \times I} \rightarrow \pi^*(E_2)|_{A \times I} , \quad \Psi(\xi_{x,t}) := ((x, t), \Phi_t(\xi_x))$$

This is a bundle isomorphism, since  $\Phi_t$  is one. Now denote the inclusion

$$i_t : X \rightarrow X \times I, \quad i_t(x) := (x, t) ,$$

and notice that the pullback along  $i_t$  is the same as restricting to  $X \times \{t\}$ .

With those definitions it is clear by the construction of the gluing, that

$$E_1 \cup_{\Phi_t} E_2 \cong i_t^* (\pi^*(E_1) \cup_\Psi \pi^*(E_2)) . \quad (1)$$

{eq:kTheo-glu  
ingHomotopyIn  
variant}

The isomorphism can be seen by part (2) of Corollary 3.2.10. as follows: First notice the isomorphism  $\pi^*(E_i)|_{A \times \{t\}} \cong E_i$  for  $i = 1, 2$  and any  $t$ . By construction of the gluing this leads to the isomorphism

$$E_1 \cup_{\Phi_t} E_2 \cong \pi^*(E_1)|_{A \times \{t\}} \cup_{(\Phi_t, \text{id})} \pi^*(E_2)|_{A \times \{t\}} .$$

Now again by construction we notice, that gluing restricted spaces is the same as gluing the whole space and then restricting. Furthermore the map  $(\Phi_t, \text{id})$  is the restriction of  $\Psi$  to  $X \times \{t\}$ . Therefore:

$$\pi^*(E_1)|_{A \times \{t\}} \cup_{(\Phi_t, \text{id})} \pi^*(E_2)|_{A \times \{t\}} \cong (\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X \times \{t\}}$$

Finally, pulling back via  $i_t$  is the same as restricting  $X \times i$  to  $X \times \{t\}$  and identifying  $X$  with  $X \times \{t\}$ . This concludes the chain of arguments:

$$(\pi^*(E_1) \cup_\Psi \pi^*(E_2))|_{X \times \{t\}} \cong i_t^* (\pi^*(E_1) \cup_\Psi \pi^*(E_2)) .$$

Since the pullback is homotopy invariant we can conclude by the isomorphism (1):

$$E_1 \cup_{\Phi_0} E_2 \cong E_1 \cup_{\Phi_1} E_2 .$$

□

def:kTheo-hom  
otopyClasses0  
fMaps

**Definition 3.2.12.** Let  $X, Y$  be topological spaces. By  $[X, Y]$  we denote the set of homotopy classes of maps from  $X$  to  $Y$ .



lem:kTheo-gluingAndHomotopyClassesToGLn

**Lemma 3.2.13.** *Let  $X = X_1 \cup X_2$  with  $X_1$  and  $X_2$  closed contractible spaces. Then there is a canonical bijection*

$$\text{Vect}_n(X) \rightarrow [A, \text{GL}_n(\mathbb{C})].$$

*Proof.* Let  $[E]$  be any vector bundle over  $X$  of rank  $n$ . Then  $E$  can be represented by a vector bundle obtained by gluing that is trivial on both  $X_i$ : Since  $X_i$  is contractible we can choose two trivializations

$$\alpha_1 : E|_{X_1} \rightarrow X_1 \times \mathbb{C}^n, \quad \text{and} \quad \alpha_2 : E|_{X_2} \rightarrow X_2 \times \mathbb{C}^n.$$

With part (2) of Corollary 3.2.10 we can conclude:

$$E \cong E|_{X_1} \cup_{\text{id}_A} E|_{X_2} \cong X \times \mathbb{C}^n \cup_{\alpha_2 \circ \alpha_1^{-1}|_A} X \times \mathbb{C}^n$$

Since a map between the two trivial bundles  $A \times \mathbb{C}^n \rightarrow A \times \mathbb{C}^n$  can be identified with a map  $A \rightarrow \text{GL}_n(\mathbb{C})$  we get an assignment from bundles over  $X$  together with the trivializations to maps  $A \rightarrow \text{GL}_n(\mathbb{C})$ . The next step is to show that two isomorphic vector bundles give homotopic maps: Assume that we are given two vector bundles  $E, F$  over  $X$  that are isomorphic via  $T : E \rightarrow F$  and without loss of generality we assume that they are given as the gluing of two trivial bundles:

$$E : X_1 \times \mathbb{C}^n \cup_{\varphi} X_2 \times \mathbb{C}^n, \quad \text{and} \quad F : X_1 \times \mathbb{C}^n \cup_{\psi} X_2 \times \mathbb{C}^n.$$

We now want to define a homotopy between  $\psi$  and  $\varphi$ . For this we start by splitting the isomorphism  $T$  into its two parts over  $X_1$  and  $X_2$ :

$$T|_{X_i} : X_i \times \mathbb{C}^n \rightarrow X_i \times \mathbb{C}^n.$$

These maps are given by maps

$$T_i : X_i \rightarrow \text{GL}_n(\mathbb{C}),$$

where the gluing compatibility of the  $T|_{X_i}$  yield for any  $a \in A$ :

$$\psi(a)T_1(a) = T_2(a)\varphi(a), \quad \text{equivalently} \quad \psi(a) = T_2(a)\varphi(a)T_1^{-1}(a).$$

By the contractibility of  $X_i$  we have for any fixed  $x_i \in X_i$  a homotopy  $h_t^i : X_i \rightarrow X_i$  such that  $h_0^i = c_{x_i}$  is the collapsing map whilst  $h_1^i = \text{id}_{X_i}$ . Furthermore, since  $\text{GL}_n(\mathbb{C})$  is connected we have a path  $\gamma^i : I \rightarrow \text{GL}_n(\mathbb{C})$  that connects  $T_i(x_i) = \gamma^i(1)$  to  $\mathbb{1}_n = \gamma^i(0)$ . With this we can define the homotopies

$$T_{i,t} : X_i \rightarrow \text{GL}_n(\mathbb{C})$$

$$x \mapsto \begin{cases} T_i \circ h_{2t}^i(x) & \text{if } t \leq \frac{1}{2}, \\ \gamma^i(2(t - \frac{1}{2})) & \text{if } t > \frac{1}{2}, \end{cases}$$

that satisfy  $T_{i,1} = T_i$  and  $T_{i,0}(x) = \mathbb{1}_n$  for all  $x$ . Finally we define the homotopy  $H_t : A \rightarrow \mathrm{GL}_n(\mathbb{C})$  by

$$H_t(a) := T_{2,t}(a)\varphi(a)T_{1,t}^{-1}(a)$$

to get a homotopy from  $\varphi = H_0$  to  $\psi = H_1$ . Hence, we get a well defined map from isomorphism classes of bundles over  $X$  of rank  $n$  to the homotopy classes of maps from  $A$  to  $\mathrm{GL}_n(\mathbb{C})$  that is the inverse of the gluing construction and hence is bijective.  $\square$

def:kTheo-met  
ricsOnBundles

**Definition 3.2.14** (Metrics on Bundles). A **metric** on a complex bundle  $E$  is a section  $h : X \rightarrow \mathrm{Herm}(E)$  such that  $h(x)$  is positive definite for all  $x$ .  $\mathrm{Herm}(E)$  denotes the bundle of hermitian forms (conjugate-symmetric sesquilinear) in each fiber. A bundle together with a metric is a **Hermitian bundle**.

def:kTheo-pro  
jectionOperat  
or

**Definition 3.2.15** (Projection Operator). A **projection operator**  $P : E \rightarrow E$  is a bundlehomomorphism in  $E$  such that  $P^2 = P$ .

lem:kTheo-pro  
jectionsGives  
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on

**Lemma 3.2.16.** *Any projection operator  $P : E \rightarrow E$  determines a direct sum decomposition  $E = PE \oplus (1 - P)E$ .*

*Proof.* Observe that fibrewise  $1 - P$  maps each  $\xi$  to the kernel of  $P$ , and hence fibrewise  $P(E)$  and  $(1 - P)(E)$  are disjoint. Furthermore  $\mathrm{im}(P) + \mathrm{im}(1 - P) = E$ , because  $v = P(v) + (v - P(v))$ . Therefore in each fibre we have the direct sum  $E_x = P(E_x) \oplus (1 - P)(E_x)$ . Now the question is, if that gives a global splitting, meaning that  $P(E)$  and  $(1 - P)(E)$  are continuous as subbundles of  $E$ . For this to be true, we need local triviality, which is clear as long as the rank of  $P$  (and  $(1 - P)$ ) is locally constant. In each fiber we know that  $\mathrm{rank}(P_x) + \mathrm{rank}(1 - P) = \dim(E_x)$ . Furthermore,  $\mathrm{rank}(P_x)$  and  $\mathrm{rank}(1 - P)$  are locally constant. To see this first notice that they are lower semi continuous, meaning that for  $y$  from a small enough neighbourhood of  $x$   $\mathrm{rank}(P_x) \leq \mathrm{rank}(P_y)$ . To see this, pick a  $\mathrm{rank}(P_x)_x \times \mathrm{rank}(P_x)_x$  minor with nonzero determinant. The latter is an open condition and hence stable under small perturbation. But now we have the sum of two lower semi continuous functions that give something locally constant and hence, each summand is locally constant. Finally the above shows that  $\ker(P) = (1 - P)(E)$  we have the decomposition.  $\square$

lem:kTheo-met  
ricInducedCom  
plement

**Lemma 3.2.17** (Metric Induced Complement). *Suppose that  $F$  is a sub-bundle of  $E$ . Then a metric provides a canonical complementary sub-bundle.*

*Proof.* Let  $h$  denote the metric. For each  $x$  we have the orthonormal projection  $P_x : E_x \rightarrow F_x$  inducing a map  $P : E \rightarrow F$ . Let  $(f_1, \dots, f_n)$  be an orthonormal frame of  $F$ . Then fiberwise with  $\xi \in E_x$  the orthogonal projection is given by

$$P_x(\xi) = \sum_i h_x(\xi, f_i(x)) f_i(x).$$

And since  $h$  is continuous the whole map  $P$  is. Furthermore  $P^2 = P$  and hence we get a canonical sum decomposition.  $\square$

**Definition 3.2.18.** We call a sequence of bundle homomorphisms

$$\cdots \longrightarrow E \longrightarrow F \longrightarrow \cdots$$

exact, if it is fiberwise exact.

lem:kTheo-spl  
ittingPrinzip  
le

**Lemma 3.2.19** (Splitting Prinzip). *Every short exact sequence of bundles over the same base space splits.*

*Proof.* Let  $X$  be the base space and assume that the sequence is given as

$$0 \longrightarrow E' \xrightarrow{\varphi'} E \xrightarrow{\varphi''} E'' \longrightarrow 0$$

Now it is well known that any short exact sequence of vector spaces splits. We just need to ensure that this splitting is continuous. For this we give  $E$  a metric. Now since  $\varphi'$  is injective,  $\varphi'(E')$  is a subbundle. By Lemma 3.2.17 we can split  $E \cong \varphi'(E') \oplus L$ . However,  $\varphi''|_L$  is fiberwise (and hence as a bundle map) an isomorphism. This can be seen as  $\varphi''$  being surjective and trivial on  $\varphi'(E')$  hence the restriction is still surjective. The injectivity is also clear as the kernel by the homomorphism theorem of linear maps. But then we clearly have  $E = E' \oplus E''$  and the sequence is isomorphic as a short exact sequence to the splitting one:

$$0 \longrightarrow E' \xrightarrow{i_1} E' \oplus E'' \xrightarrow{\pi_2} E'' \longrightarrow 0$$

□

def:kTheo-amp  
leSubspace

**Definition 3.2.20.** A subspace  $V \subseteq \Gamma(E)$ , where  $\Gamma(E)$  is viewed as a  $\mathbb{C}$ -vector space is called **ample**, if

$$\varphi : X \times V \rightarrow E, \quad \varphi(x, s) := s(x)$$

is a surjection.

**Corollary 3.2.21.** If the subspace  $V$  is ample and  $\dim(V) = \text{rank}(E)$ , then  $E$  is trivializable.

lem:kTheo-amp  
leSubspaceExi  
sts

**Lemma 3.2.22.** *For any vector bundle  $E$  over a compact Hausdorff space  $X$ , there exists an finite dimensional ample subspace of  $\Gamma(E)$ .*

*Proof.* Let  $\{U_\alpha\}$  be a finite open cover of  $X$  such that  $E|_{U_\alpha}$  is trivial. Let  $\{s_{\alpha,i}\}$  be the corresponding local frames, and define  $V_\alpha$  to be the finite dimensional ample subspace  $V_\alpha \subseteq \Gamma(E|_{U_\alpha})$ . Let  $\rho_\alpha$  be a partition of unity subordinate to the cover  $\{U_\alpha\}$ . Define

$$\theta_\alpha : V_\alpha \rightarrow \Gamma(E), \quad \theta_\alpha(s_\alpha)(x) = \begin{cases} s_\alpha(x) \cdot \rho_\alpha(x) & \text{if } x \in U_\alpha, \\ 0 & \text{else.} \end{cases}$$

Then the  $\theta_\alpha$  give rise to a homomorphism

$$\theta : \Pi_\alpha V_\alpha \rightarrow \Gamma(E),$$

and the image of  $\theta$  is finite dimensional and ample. The surjectivity is clear, as for any  $x$  there exists a  $\rho_\alpha(x) > 0$  making the map

$$\theta_\alpha(V_\alpha) \rightarrow E_x$$

a surjection.  $\square$

cor:kTheo-epi  
FromTrivial

**Corollary 3.2.23.** If  $E$  is any bundle over  $X$ , there exists an epimorphism  $\varphi : X \times \mathbb{C}^m \rightarrow E$  for some integer  $m$ .

*Proof.* Just take a basis  $(s_1, \dots, s_m)$  of an ample subspace  $V$  and define  $\varphi(x, e_i) = s_i(x)$ , where  $e_i$  is the  $i$ -th standard basis vector of  $\mathbb{C}^m$ .  $\square$

cor:kTheo-spl  
ittingTrivial

**Corollary 3.2.24.** If  $E$  is a bundle over  $X$ , there exists a bundle  $F$  such that  $E \oplus F$  is trivial.

*Proof.* By Corollary 3.2.23 there exists an epimorphism  $\varphi : X \times \mathbb{C}^m \rightarrow E$ . The kernel of  $\varphi$  is a subbundle of  $X \times \mathbb{C}^m$  and hence by Lemma 3.2.19 we have a splitting of the short exact sequence

$$0 \longrightarrow \ker(\varphi) \longrightarrow X \times \mathbb{C}^m \xrightarrow{\varphi} E \longrightarrow 0$$

Hence, the middle term becomes isomorphic to the direct sum of the outer terms and we get  $X \times \mathbb{C}^m \cong E \oplus \ker(\varphi)$ .  $\square$

def:kTheo-gra  
ssmanManifold

**Definition 3.2.25** (Grassmann Manifold). Let  $V$  be a vector space and  $n$  be any integer. The **Grassmann manifold**  $\text{Gr}_n(V)$  is the set of all subspaces of  $V$  with **codimension**  $n$ . If  $V$  is given a Hermitian metric, we can topologize  $\text{Gr}_n(V)$  as a subspace of the set of all projection operators on  $V$  and this defines a map

$$P : \text{Gr}_n(V) \rightarrow \text{End}(V),$$

where  $\text{End}(V)$  is the space of all endomorphisms of  $V$  with the operator norm topology. We give  $\text{Gr}_n(V)$  the subspace topology of  $\text{End}(V)$ .

def:kTheo-cla  
ssifyingMap

**Definition 3.2.26.** Suppose that  $E$  is a bundle over  $X$ ,  $V$  is a vector space and  $\varphi : X \times V \rightarrow E$  is an epimorphism. Then we can define a map  $f : X \rightarrow \text{Gr}_{\dim(V)}(V)$  by sending  $x$  to the kernel of  $\varphi_x : V \rightarrow E_x$ . We call  $f$  to be induced by  $\varphi$ . This map is continuous for any metric on  $V$ . To see this notice that the topology on  $V$  is metric invariant, as  $V$  is finite dimensional. Now the map  $f$  is continuous if the map for the map

$$P : \text{Gr}_n(V) \rightarrow \text{End}(V)$$

the composition  $P \circ f$  is continuous. But  $P \circ f$  is given by sending  $x$  to the projection onto  $\ker(\varphi_x)$ , which is continuous as  $\varphi$  is.

def:kTheo-cla  
ssifyingBundl  
e

**Definition 3.2.27** (Classifying Bundle). Let  $V$  be a vector space and  $F \subseteq \text{Gr}_n(V) \times V$  be the subbundle consisting of all points  $(g, v)$  such that  $v \in g$ . We call  $F$  the **tautological bundle** over  $\text{Gr}_n(V)$ . Furthermore, if  $E = (\text{Gr}_n(V) \times V)/F$  is the quotient bundle, we call  $E$  the **classifying bundle** over  $\text{Gr}_n(V)$ .

**Corollary 3.2.28.** If  $E'$  is a bundle over  $X$ , and  $\varphi : X \times V \rightarrow E'$  is an epimorphism, then  $\varphi$  induces a map  $f : X \rightarrow \text{Gr}_m(V)$  such that  $E' \cong f^*E$ , where  $E$  is the classifying bundle over  $\text{Gr}_m(V)$ . The isomorphism follows from a fiberwise construction because  $E_x \cong V / \ker(\varphi_x)$ . The latter however is exactly the fiber over  $\ker(x)$  in the classifying bundle and hence the pullback via the  $\varphi$  induced map gives an isomorphism. That motivates the name "classifying bundle" for  $E$  and "classifying map" for  $f$ .

lem:kTheo-Gra  
ssmanTopology  
MetricInvaria  
nt

**Lemma 3.2.29.** *The topology on  $\text{Gr}_n(V)$  is independent of the choice of the metric on  $V$ .*

*Proof.* Let  $h$  and  $h'$  be two metrics on  $V$ . Then we denote the topological space  $\text{Gr}_n(V)$  with the topology induced by  $h$  by  $\text{Gr}(V_h)$  and the one induced by  $h'$  by  $\text{Gr}(V_{h'})$ . We need to show that the identity map  $\text{id} : \text{Gr}(V_h) \rightarrow \text{Gr}(V_{h'})$  is a homeomorphism. For this we need to show that  $\text{id}$  is continuous. The projection  $\text{Gr}_n(V_h) \times V_{h'} \rightarrow E$  (where  $E$  is the classifying bundle) is an epimorphism and hence induces a classifying map  $f : \text{Gr}_n(V_h) \rightarrow \text{Gr}_n(V_{h'})$  such that  $f^*E \cong E$ . However that map is given by the identity on  $\text{Gr}_n(V)$  and hence the identity is continuous by Definition 3.2.26.  $\square$

**Corollary 3.2.30.** The inverse system given by the projections

$$\pi : \mathbb{C}^m \rightarrow \mathbb{C}^{m-1}, \quad (x_1, \dots, x_m) \mapsto (x_1, \dots, x_{m-1}),$$

induces an direct system of Grassmann manifolds with the induced continuous maps

$$i_{m-1} : \text{Gr}_n(\mathbb{C}^{m-1}) \rightarrow \text{Gr}_n(\mathbb{C}^m), \quad U \mapsto \pi^{-1}(U).$$

If we denote the classifying bundle over  $\text{Gr}_n(\mathbb{C}^m)$  by  $E_m$ , then the pullback of  $E_m$  along  $i_{m-1}$  is isomorphic to  $E_{m-1}$ .

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**Theorem 3.2.31.** *The map*

$$\lim_{m \rightarrow \infty} [X, \text{Gr}_n(\mathbb{C}^m)] \rightarrow \text{Vect}_n(X),$$

*induced by  $f \mapsto f^*E_m$  is a bijection for all compact Hausdorff spaces  $X$ .*

*Proof.* We shall construct an inverse map. If  $E$  is a bundle over  $X$ , then by Corollary 3.2.23 there exists an epimorphism  $\varphi : X \times \mathbb{C}^m \rightarrow E$  for some integer  $m$ . Let  $f : X \rightarrow \text{Gr}_n(\mathbb{C}^m)$  be the map induced by  $\varphi$ . If we manage to show that the homotopy class of

$f$  as a map to  $\mathrm{Gr}_n(\mathbb{C}^{m'})$  for some large enough  $m'$  is independent of the choice of  $\varphi$  and  $m$ , then we get a well defined map from  $\mathrm{Vect}_n(X)$  to  $\lim_{m \rightarrow \infty} [X, \mathrm{Gr}_n(\mathbb{C}^m)]$  that is inverse to the classifying map.

Suppose that  $\varphi_i : X \times \mathbb{C}^{m_i} \rightarrow E$  for  $i = 1, 2$  are two epimorphisms. Let  $g_i : X \rightarrow \mathrm{Gr}_n(\mathbb{C}^{m_i})$  be the maps induced by  $\varphi_i$ . We want to show that  $g_1$  and  $g_2$  are homotopic as maps to  $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$ . For this we define the map

$$\psi_t : X \times \mathbb{C}^{m_1} \times \mathbb{C}^{m_2} \rightarrow E, \quad \psi_t(x, v_1, v_2) = t\varphi_1(x, v_1) + (1-t)\varphi_2(x, v_2).$$

Since the  $\varphi_i$  are epimorphisms, this map is an epimorphism for any  $t$ . Let

$$f_t : X \rightarrow \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$$

be the maps induced by  $\psi_t$ . (where we identified  $\mathbb{C}^{m_1} \times \mathbb{C}^{m_2}$  with  $\mathbb{C}^{m_1+m_2}$  by mapping  $(v_1, v_2)$  to  $(v_1, v_2)$ ). Then

$$f_0 = j_0 \circ g_1 \quad \text{and} \quad f_1 = T \circ j_1 \circ g_2,$$

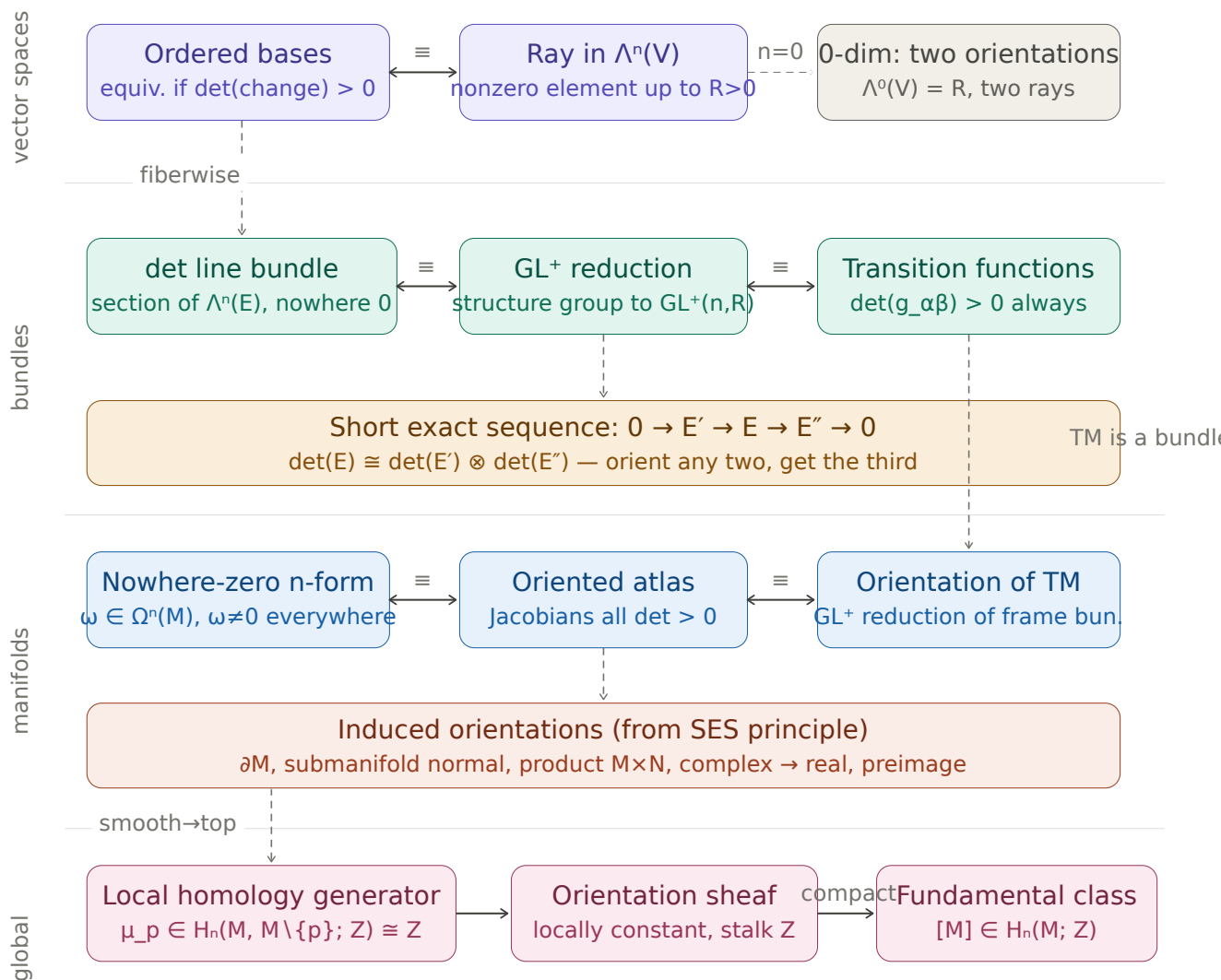
where  $j_i : \mathrm{Gr}_n(\mathbb{C}^{m_i}) \rightarrow \mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$  is the map induced by the inclusion  $\mathbb{C}^{m_i} \rightarrow \mathbb{C}^{m_1+m_2}$  and  $T$  is the map induced by the isomorphism  $\mathbb{C}^{m_1+m_2} \rightarrow \mathbb{C}^{m_1+m_2}$  that sends  $(v_1, v_2)$  to  $(v_2, v_1)$ . Since  $T$  is homotopic to the identity, we can conclude that  $j_0 \circ g_1$  and  $j_1 \circ g_2$  are homotopic as maps to  $\mathrm{Gr}_n(\mathbb{C}^{m_1+m_2})$ . Hence, the homotopy class of the map induced by an epimorphism is independent of the choice of the epimorphism and the integer  $m$ . This concludes the proof.  $\square$

## 4 Orientations

### 4.1 Mathematical Goals

**Theorem 4.1.1** (The Theorems and Mathematical Structures). • *Define Orientations of Manifolds*

• *How to induce Orientations*



$[M] \cap - : H^k(M) \rightarrow H_{n-k}(M)$  (Poincaré dual)

The approach to this chapter is ...



## 5 Morse Theory

For everything concerning Morse theory our main reference will be the wonderful book [BH04]. In Morse theory, we will always deal with a smooth function  $f : M \rightarrow \mathbb{R}$  from a smooth Riemannian manifold  $(M, g)$  that is also compact. We will denote its dimension by  $m = \dim(M)$ .

**Definition 5.0.1** (Critical Points). A point  $p \in M$  at which the smooth map  $df_p = 0$  is called **critical**. By  $\text{Crit}(f)$  we denote the **set of critical points**.

**Definition 5.0.2** (The Hessian). Let  $f : M \rightarrow \mathbb{R}$  be a smooth function. The **Hessian** of  $f$  at  $p \in \text{Crit}(f)$  is a symmetric bilinear form

$$\text{Hess}_p(f) : T_p M \times T_p M \rightarrow \mathbb{R}$$

that is defined as follows. Let  $\xi_p, \zeta_p \in T_p M$  be two tangent vectors. Now choose two vector fields  $\Xi$  and  $Z$  in a neighbourhood of  $p$  such that  $\Xi(p) = \xi_p$  and  $Z(p) = \zeta_p$ . We then define the Hessian to be:

$$\text{Hess}_p(f)(\xi_p, \zeta_p) = \Xi(Z(f))(p)$$

See 2.1.33 for the notation of applying a function to a vector field. In corollary 5.0.5 we will give a local description telling us that this definition is well defined, meaning independent of the Extensions  $\Xi$  and  $Z$ .

**Definition 5.0.3.** A smooth function  $f : M \rightarrow \mathbb{R}$  is called **Morse**, if for any critical point, the Hessian is a symmetric bilinear and non-degenerate form.

For such a form there exists the following theorem:

**Theorem 5.0.4** (Sylvester's Law of Inertia). *Let  $A \in \text{Mat}(n, \mathbb{R})$  be symmetric and let  $T, T' \in \text{GL}(n, \mathbb{R})$  and  $k, k', l, l' \in \mathbb{N}$  such that:*

$$T^\top \circ A \circ T = \begin{pmatrix} \mathbb{1}_k & 0 & 0 \\ 0 & -\mathbb{1}_l & 0 \\ 0 & 0 & 0_{n-k-l} \end{pmatrix} \text{ and } T'^\top \circ A \circ T' = \begin{pmatrix} \mathbb{1}_{k'} & 0 & 0 \\ 0 & -\mathbb{1}_{l'} & 0 \\ 0 & 0 & 0_{n-k'-l'} \end{pmatrix}$$

*then  $k = k'$ ,  $l = l'$  and  $\text{rank}(A) = k + l$ .*

*Proof.* Since  $T, T'$  are invertible we have that

$$k + l = \text{rank}(T^\top \circ A \circ T) = \text{rank}(A) = \text{rank}(T'^\top \circ A \circ T') = k' + l'. \quad (2)$$

Hence it suffices to show that  $k = k'$ . Which we will do by proving the claim:

$$k = \max \{ \dim(U) \mid U \subseteq \mathbb{R}^n \text{ subspace such that } x^\top A x > 0 \ \forall x \in U \setminus \{0\} \}$$

So we start by showing “ $\leq$ ” : Denote the first  $k$  columns of  $T$  with  $x_1, \dots, x_k$ . They form a basis of the subspace  $\mathbb{R}^k \subseteq \mathbb{R}^n$  and with  $0 \neq x = \sum_{i=1}^k \lambda_i x_i$  we have that by bilinearity

$$x^\top A x = \sum_{i=1}^k \lambda_i x_i^\top A x = \sum_{i,j=1}^k \lambda_i \lambda_j x_i^\top A x_j = \sum_{i=1}^k (\lambda_i)^2 \geq 0. \quad (3)$$

This concludes the first inequality.

Now let  $U$  be any  $k$ -dimensional subspace such that for all non zero  $x \in U$ ,  $x^\top A x > 0$ . By a calculation analogous to the one above we have that for any  $x \in W := \text{span}(x_{k+1}, \dots, x_n)$  the number  $x^\top A x$  is less than or equal to zero. Hence,  $W \cap U = \{0\}$  and with this we conclude:

$$\dim(U) = \dim(U + W) - \dim(W) + \dim(U \cap W) \leq (k + (n - k)) - (n - k) + 0 = k. \quad (4)$$

This completes the proof.  $\square$

cor:morse-loc  
alHessian

**Corollary 5.0.5.** Let  $\varphi : U \rightarrow V$  be a chart around a critical point  $p \in M$ . Then in said chart the matrix corresponding to the Hessian has the form:

$$M_p(f) = \left( \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j} \right)_{i,j}$$

*Proof.* This follows by direct calculation of  $\text{Hess}_p(f)(\partial_i(p), \partial_j(p))$ . Denote  $x := \varphi(p)$ , then:

$$\begin{aligned} (\partial_i(p) (\partial_j(f)))_p &= \left( \partial_i(p) \left( p \mapsto \frac{\partial f \circ \varphi^{-1}}{\partial x_j} \Big|_{\varphi(p)} \right)_p \right) \\ &= \frac{\partial}{\partial x_i} \left( y \mapsto \frac{\partial f \circ \varphi^{-1}}{\partial x_j} \Big|_y \right) \Big|_x \\ &= \frac{\partial^2 (f \circ \varphi^{-1})}{\partial x_i \partial x_j}. \end{aligned}$$

$\square$

**Remark 5.0.6.** If we have two different charts, then the matrix  $M_p(f)$  of the Hessian transforms from one basis to the other via conjugation of the differential of the transition function  $\varphi_{12} := \varphi_1 \circ \varphi_2^{-1}$ :

$$M_p^1(f) = (d\varphi_{12})^\top \cdot M_p^2(f) \cdot d\varphi_{12}$$

Hence we can define the following:

**Definition 5.0.7.** By Sylvester’s Law of inertia, the number  $l$  that denotes the dimension of the maximal subspace on which a matrix is negative definite, is an invariant under conjugation. We call that number the index of said matrix. Now in the setting of Morse theory we assign to each critical point  $p$  the index of  $M_p(f)$  and call that the **(Morse-)index** of  $p$ , denoted by  $\lambda_p$ . If omitting the function would cause ambiguity, we write  $\lambda_p^f$ . We denote the set of all critical points of index  $k$  by  $\text{Crit}_k(f)$ .

thm:morse-morseLemma

**Theorem 5.0.8** (Morse Lemma). *Let  $p$  be a critical point of index  $k$  of the Morse function  $f : M \rightarrow \mathbb{R}$ . There exists a centered chart  $\varphi : U \rightarrow \mathbb{R}^m$  around  $p$  such that*

$$(f \circ \varphi^{-1})(x_1, \dots, x_m) = f(p) + \sum_{i=1}^k -x_i^2 + \sum_{i=k+1}^m x_i^2.$$

*Proof.* We refer to Lemma 3.11 in [\[banyaga2004lectures\]](#) for a proof. □

6 Title

6.1 Mathematical Goals

**Theorem 6.1.1** (The Theorems and Mathematical Structures).

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The approach to this chapter is ...

7 Title

7.1 Mathematical Goals

- Theorem 7.1.1** (The Theorems and Mathematical Structures).

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